

Automatic A/C Circuit — Knowledge Document

****Vehicle:**** 2020 Jeep Compass Altitude (2.4L Eng)
****Source:**** ProDemand wiring diagram manual, "AIR CONDITIONING" section,
Automatic A/C Circuit Fig. 1-5 of 5
****Related but distinct circuit:**** ***Manual A/C Circuit*** (Fig. 6-10 of 5 in the same manual) — same power/relay layout but omits the Sun Sensor, In-Car Temperature Sensor, and Auto-Headlamp signal, since a manually-operated system doesn't need those automatic-climate-control inputs.

> ****Note on accuracy:**** This document was compiled from an OCR'd scan of a multi-page fold-out wiring diagram. Wire colors, component names, connector/splice IDs, and general signal flow are transcribed as printed. Exact pin numbers on the 30-way HVAC Module connector could not be perfectly aligned from the scan and should be treated as approximate — always confirm exact pin location against the original diagram or a factory pinout tool before probing/repairing.

1. System Overview

The Automatic A/C system is built around the ****HVAC Module**** (center of dash), which:

- Is powered by the ****Blower Motor Relay**** and ****A/C Clutch Relay**** (both inside the Power Distribution Center).
- Talks to the rest of the vehicle over ****CAN IHS**** (interior high-speed bus).
- Reads six temperature sensors, a sun-load sensor, and door position feedback to automatically drive three blend/mode/recirculation doors and the blower motor.
- Commands the A/C compressor clutch, which is ultimately grounded/switched through the ****Powertrain Control Module (PCM)**** and the ****A/C Compressor Solenoid****, with pressure feedback from the ****A/C Pressure Transducer**** and a hard interlock via the ****Series/Parallel Radiator Fan Relay****.

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 PDC["Power Distribution Center
Fuse 83 (40A) / Fuse 19 (7.5A)"] -> BMR["Blower Motor Relay"]
 PDC --> ACR["A/C Clutch Relay"]
 BMR --> FBMPM["Front Blower Motor
Power Module"]
 FBMPM --> BLWR["Blower Motor"]
 ACR --> HVAC["HVAC Module
(Center of Dash)"]
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HVAC <-- CAN IHS --> BCM["Body Control Module"]
HVAC -- Blend/Mode/Recirc control --> LBD["Left Blend Door Actuator"]
HVAC --> RBD["Right Blend Door Actuator"]
HVAC --> MDA["Front Mode Door Actuator"]
HVAC --> RDA["Recirculation Door Actuator"]
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ICT["In-Car Temp Sensor"] --> HVAC
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SUN["Sun Sensor"] --> HVAC
AAT["Ambient Air Temp Sensor"] --> BCM --> HVAC
EVAP["Evaporator Temp Sensor"] --> HVAC
DUCTS["4x Duct Temp Sensors
(LH/RH Panel, LH/RH Floor)"] --> BCM
--> HVAC

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HVAC -- A/C Clutch Ctrl O/P --> PCM["Powertrain Control Module"]
PCM -- A/C Compressor Ctrl --> ACSOL["A/C Compressor Solenoid"]
ACSOL --> ACCLUTCH["A/C Compressor Clutch"]
APT["A/C Pressure Transducer"] --> PCM
ECT["Engine Coolant Temp Sensor"] --> PCM
PCM <-- CAN C --> RFM["Radiator Fan Module"]
HUMID["Humidity Sensor
(in mirror)"] -- LIN Bus --> BCM
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## ## 2. Power Supply & Fuses

| Fuse | Rating | Location                                              | Feeds                                                                                 |
|------|--------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| 83   | 40A    | Power Distribution Center Assembly                    | Blower Motor Relay (pin 30)                                                           |
| 19   | 7.5A   | Power Distribution Center Assembly                    | A/C Clutch Relay (pin 30)                                                             |
| 1    | 70A    | Power Distribution Center Assembly (hot at all times) | Fused B(+) to Body Control Module circuit board                                       |
| 36   | 20A    | Power Distribution Center Assembly                    | Body Control Module circuit board – ambient/duct temp sensor supply, hood ajar switch |
| 37   | 10A    | Power Distribution Center Assembly                    | Body Control Module circuit board                                                     |
| 51   | 7.5A   | Power Distribution Center Assembly                    | Fused ignition (run/start) to Body Control Module circuit board                       |
| 9    | 7.5A   | Power Distribution Center Assembly                    | Series/Parallel Radiator Fan Relay, A/C Compressor Solenoid circuit                   |
| A    | 50A    | Battery Distribution Unit                             | Radiator Fan Module (FSD B+)                                                          |

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## ## 3. Relays

| Relay                              | Location                                              | Control (coil)                              | Switched output                                                           |
|------------------------------------|-------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------|
| Blower Motor Relay                 | Power Distribution Center                             | Driven by HVAC Module (BLWR CTRL)           | Fuse 83 (40A) → Front Blower Motor Power Module                           |
| A/C Clutch Relay                   | Power Distribution Center                             | Driven by HVAC Module (A/C Clutch Ctrl O/P) | Fuse 19 (7.5A) → A/C clutch control feed                                  |
| Series/Parallel Radiator Fan Relay | Power Distribution Center (left side of engine compt) | Driven by Radiator Fan Module / PCM         | Fuse 9 (7.5A) circuit, gates A/C compressor clutch & radiator fan staging |

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## ## 4. HVAC Module – Signal Reference

The HVAC Module is a 30-way connector (plus a small 6-way ground/splice connector, `C1`, via splice `GZ913A`/`GZ913B`) located at the center of the dash. Signals present at this connector:

| Signal                          | Wire Color       | Destination                                                    |
|---------------------------------|------------------|----------------------------------------------------------------|
| Fused Ignition                  | PNK/GRN          | Fuse 51 (via PDC)                                              |
| Fused B(+)                      | YEL/RED          | Fuse 19/Battery feed                                           |
| CAN IHS (+)                     | YEL/GRY          | Computer Data Lines System                                     |
| CAN IHS (-)                     | WHT/PNK          | Computer Data Lines System                                     |
| Ground                          | BLK/VIO          | Splice `Z913` / `GZ913A` / `GZ913B` (left side of engine comp) |
| LH Blend Door Common Drive      | WHT/RED          | Left Blend Door Actuator                                       |
| LH Blend Door Drive             | ORG/BLK          | Left Blend Door Actuator                                       |
| LH Blend Door Signal            | BLU/WHT          | Left Blend Door Actuator (position feedback)                   |
| A/C Clutch Control Output       | RED/GRN          | A/C Clutch Relay coil                                          |
| Blower Control (BLWR CTRL)      | BLU/GRN          | Blower Motor Relay coil                                        |
| Blower Motor Return             | BLU/BLK          | Front Blower Motor Power Module                                |
| Mode Door Drive                 | RED/PNK          | Front Mode Door Actuator                                       |
| Mode Door Common Drive          | GRY/BLK          | Front Mode Door Actuator                                       |
| Mode Door Signal                | GRY/GRN          | Front Mode Door Actuator (position feedback)                   |
| RH Panel Temp Sensor Signal     | BRN              | Right Panel Duct Temperature Sensor                            |
| LH Panel Temp Sensor Signal     | GRN              | Left Panel Duct Temperature Sensor                             |
| Evaporator Temp Sensor Signal   | BLU/YEL          | Evaporator Temperature Sensor                                  |
| Reference Voltage Supply        | WHT/GRN          | Shared sensor excitation supply                                |
| Recirculation Door Common Drive | GRY/RED          | Recirculation Door Actuator                                    |
| Recirculation Door Drive        | BRN/GRN          | Recirculation Door Actuator                                    |
| Recirculation Door Signal       | BRN/BLK          | Recirculation Door Actuator (position feedback)                |
| RH Floor Temp Sensor Signal     | WHT              | Right Floor Duct Temperature Sensor                            |
| LH Floor Temp Sensor Signal     | ORG              | Left Floor Duct Temperature Sensor                             |
| Sensor Ground (x2)              | BLK/RED, BLK/VIO | Shared sensor return                                           |
| RH Blend Door Signal            | WHT/GRY          | Right Blend Door Actuator                                      |
| RH Blend Door Drive             | YEL/GRY          | Right Blend Door Actuator                                      |
| RH Blend Door Common Drive      | GRN/BLK          | Right Blend Door Actuator                                      |
| Auto Headlamp Signal            | WHT/VIO          | Body Control Module (headlamp switch logic)                    |
| Sun Sensor Signal 1             | VIO/DK BLU       | Sun Sensor                                                     |
| Sun Sensor Signal 2             | VIO/BLU          | Sun Sensor                                                     |
| In-Car Temperature Signal       | BLU/ORG          | In-Car Temperature Sensor                                      |
| In-Car Temperature Return       | BLU/BRN          | In-Car Temperature Sensor                                      |

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## ## 5. Temperature & Position Sensors

| Sensor                                  | Location                          | Signal(s)                                | Routed to                                                         |
|-----------------------------------------|-----------------------------------|------------------------------------------|-------------------------------------------------------------------|
| In-Car Temperature Sensor               | Under left side of dash           | IN CAR TEMP SIG, IN CAR TEMP RTN, GND    | HVAC Module (direct)                                              |
| Sun Sensor                              | Front center of dash              | SUN SENS SIG 1, SUN SENS SIG 2, SENS GND | HVAC Module (direct)                                              |
| Ambient Air Temperature Sensor          | Behind right side of front fascia | AAT SIG, SENS GND                        | Body Control Module → HVAC Module                                 |
| Evaporator Temperature Sensor           | Left side of HVAC unit            | EVAP TEMP SENS SIG, SENS GND             | HVAC Module (direct), also shown routed through BCM circuit board |
| Left Panel Duct Temperature Sensor      | Center of dash                    | SENS SIG, SENS GND                       | HVAC Module (via BCM circuit board)                               |
| Right Panel Duct Temperature Sensor     | Right side of dash                | SENS SIG, SENS GND                       | HVAC Module (via BCM circuit board)                               |
| Left Floor Duct Temperature Sensor      | Center of dash                    | SENS SIG, SENS GND                       | HVAC Module (via BCM circuit board)                               |
| Right Floor Duct Temperature Sensor     | Right side of dash                | SENS SIG, SENS GND                       | HVAC Module (via BCM circuit board)                               |
| Engine Coolant Temperature (ECT) Sensor | Right rear of engine              | ECT SIG, SENS GND                        | Powertrain Control Module                                         |
| A/C Pressure Transducer                 | Right front of engine comp        | A/C PRESSURE SIG, 5V SPLY, SENSOR GND    | Powertrain Control Module                                         |
| Humidity Sensor                         | Under rear-view mirror trim       | LIN BUS 1, FUSED IGN, GND                | Body Control Module (LIN bus)                                     |
| Hood Ajar Switch                        | Behind center of upper grille     | SENS SIG, SENS GND                       | Body Control Module                                               |

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## ## 6. Door Actuators & Blower

| Actuator                        | Location                | Drive signals                              | Feedback signal                    |
|---------------------------------|-------------------------|--------------------------------------------|------------------------------------|
| Left Blend Door Actuator        | Left side of HVAC       | Common Drive, Drive                        | Signal (position)                  |
| Right Blend Door Actuator       | Right side of HVAC      | Common Drive, Drive                        | Signal (position)                  |
| Front Mode Door Actuator        | Left side of HVAC       | Common Drive, Drive                        | Signal (position)                  |
| Recirculation Door Actuator     | Right side of HVAC      | Common Drive, Drive                        | Signal (position)                  |
| Blower Motor                    | Under right end of dash | BLWR MTR SPLY (×3), GND                    | —                                  |
| Front Blower Motor Power Module | Under right end of dash | BLWR CTRL (from HVAC Module), BLWR MTR RTN | Switches blower motor supply/speed |

All four door actuators share a **Reference Voltage Supply** and **Sensor Ground** pair sourced from the HVAC Module for their position potentiometers.

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## ## 7. A/C Compressor & Cooling Fan Interface

- **\*\*A/C Compressor Clutch\*\*** (mounted on the A/C compressor): grounded/energized through `DK BLU/YEL` from splice `G904` (left side of engine).
- **\*\*A/C Compressor Solenoid\*\*** (left side of engine): 3-pin - `FUSED IGN` (PNK/GRN), `LIN BUS` (WHT/BLU), `GND` (BLK) - controlled via LIN bus, gated through Fuse 9 (7.5A) and the Series/Parallel Radiator Fan Relay.
- **\*\*Radiator Fan Module\*\*** (behind left side of upper grille): supplies `RAD FAN CTRL`, `RAD FAN ENABLE SIG`, `LOW RAD FAN RLY CTRL`, `GND` to the PCM, and receives Fused B(+) from Battery Distribution Unit Fuse A (50A).
- **\*\*PCM inputs relevant to A/C\*\***: `A/C CLUTCH CTRL`, `A/C PRESSURE SIG`, `5V SPLY`, `SENSOR GND`, `ECT SIG`, plus `CAN C (+)/(-)` to the Computer Data Lines System.

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## ## 8. Key Connector / Splice / Ground Reference

| ID                      | Description                                                                        |
|-------------------------|------------------------------------------------------------------------------------|
| ---                     | ---                                                                                |
| `GZ913A` / `GZ913B`     | Splice/ground points, left side of engine compartment - HVAC Module power & ground |
| `Z913`                  | Ground point for HVAC Module                                                       |
| `G904`                  | Ground, left side of engine - A/C compressor clutch                                |
| `G939` / `G39` / `G139` | Sun sensor signal grounds/splices                                                  |
| `C1`-`C268` (various)   | HVAC Module internal connector pin groups                                          |
| `GZ905A`                | Splice, left front of engine compartment - radiator fan module                     |
| `S809`, `S778`, `S179`  | Wiring splices in dash/engine harness feeding radiator fan / A/C pressure circuits |

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## ## 9. Cross-Referenced Systems

- **\*\*Computer Data Lines System\*\*** - CAN IHS (HVAC Module ↔ BCM) and CAN C (PCM ↔ Radiator Fan Module / cluster).
- **\*\*Cooling Fans System\*\*** - Series/Parallel Radiator Fan Relay, Radiator Fan Module, Radiator Fan Resistor 1.
- **\*\*Engine Controls System\*\*** - ECT sensor, A/C pressure transducer, A/C compressor solenoid, all terminating at the PCM.
- **\*\*Body Control Module\*\*** - supplies switched power/ground to ambient and duct temperature sensors, hood ajar switch, and humidity sensor (LIN bus); relays Auto Headlamp signal to/from HVAC Module.

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## ## 10. Quick Diagnostic Notes

- No blower operation: check Fuse 83 (40A) and Blower Motor Relay before condemning the Front Blower Motor Power Module.
- No compressor clutch engagement: check Fuse 19 (7.5A) and A/C Clutch Relay at the HVAC Module side, then verify PCM `A/C CLUTCH CTRL` and A/C

Compressor Solenoid wiring (LIN bus circuit) and A/C Pressure Transducer signal (PCM won't engage clutch outside pressure range).

- Erratic auto-climate temperature control: verify In-Car Temperature Sensor and Sun Sensor circuits first (unique to Automatic A/C – absent on Manual A/C variant), then the four duct/ambient sensors routed through the BCM.

- Door not responding (blend/mode/recirc): confirm shared Reference Voltage Supply and Sensor Ground at the HVAC Module before replacing an actuator.